# Practical 2: Analysis of Nottem Time Series Dataset

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## 1. Question

The "Nottem" dataset is available in the R library and contains historical time series observations.

Using this dataset, perform the following tasks:

(a) Write the R command to load the library in which the "Nottem" dataset is available.

(b) Write the R command to load the "Nottem" dataset into the R working environment.

(c) Write the appropriate R command to view the description of the dataset and briefly explain the key characteristics of the data.

(d) Explain the nature of the data in terms of its sampling frequency.

(e) Plot the time series data and write your observations about the pattern of the data.

## 2. Objective

To explore the nottem dataset by loading it into R, investigating its metadata, and performing initial visual analysis to understand its periodicity and characteristic patterns.

## 3. R Code

```r
# (a) Load the required library
library(datasets)

# (b) Load the nottem dataset
data("nottem")

# (c) View the description of the dataset
?nottem
```

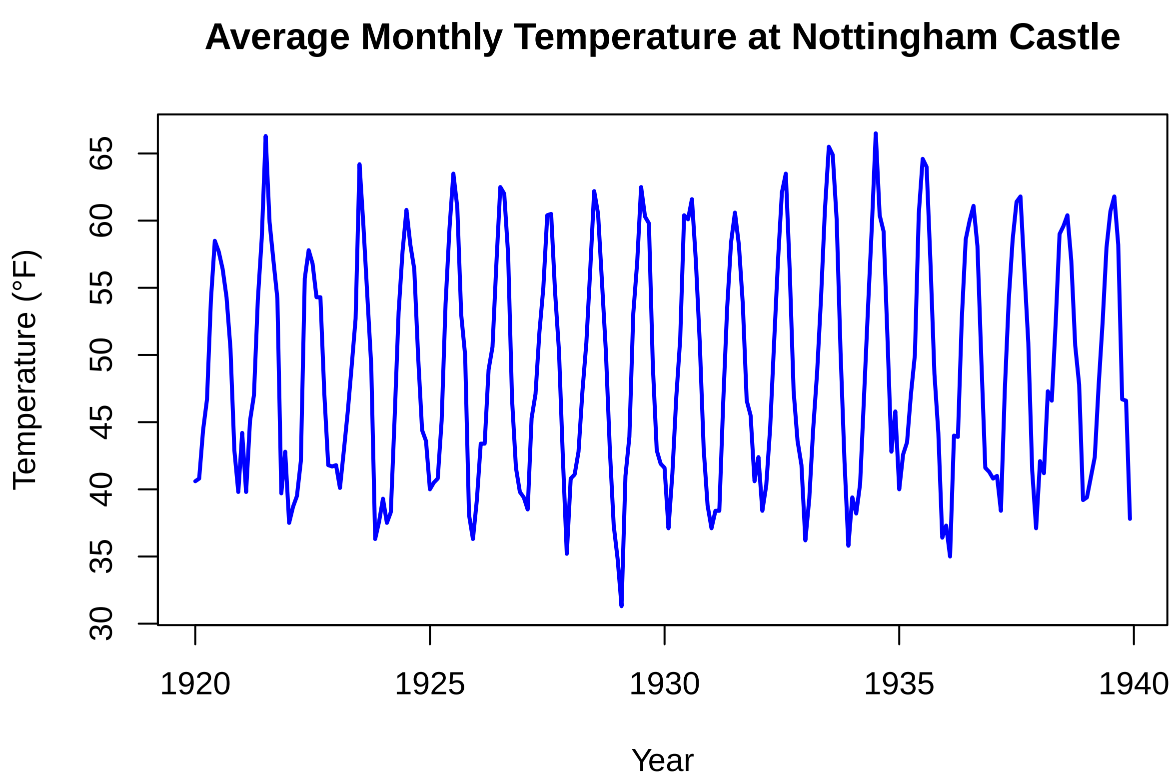
```
## starting httpd help server ... done
```

```r
# (d) Check frequency of the dataset
frequency(nottem)

# (e) Plot the dataset
plot(
  nottem,
  main = "Average Monthly Temperature at Nottingham Castle",
  xlab = "Year",
  ylab = "Temperature (°F)",
  col = "blue",
  lwd = 2
)
```

## 4. Output

### Time Series Plot




## 5. Conclusion

In this practical, we successfully loaded and visualized the nottem dataset from the datasets package in R.

From the help documentation (?nottem), we observe that the dataset contains average monthly air temperatures recorded at Nottingham Castle from 1920 to 1939.

The frequency of the dataset is 12, which indicates monthly observations. Therefore, the data is a monthly time series.

The time series plot clearly shows a strong seasonal pattern with repeating peaks and troughs every year, indicating annual climatic cycles. There is no significant long-term upward or downward trend visible during this period. Hence, the dominating component in the dataset is seasonality.